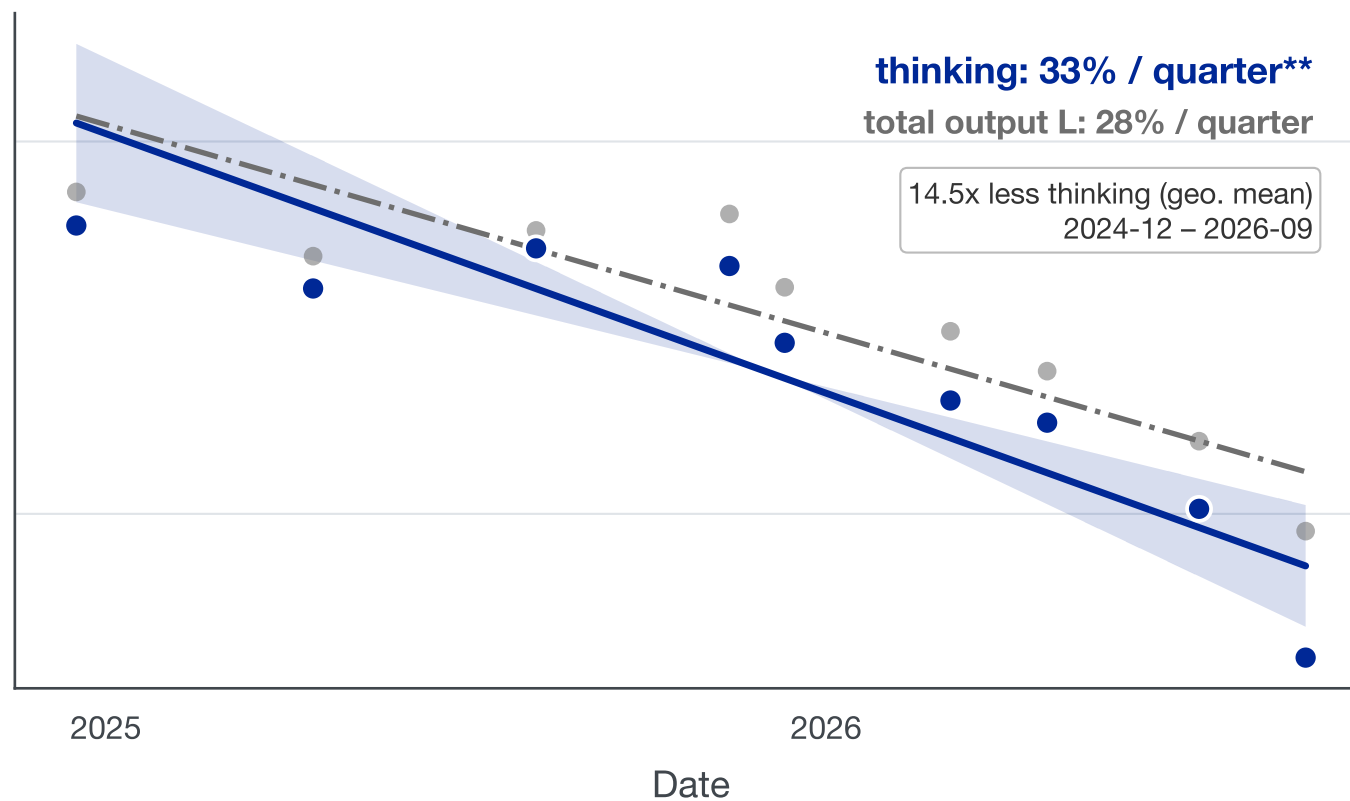


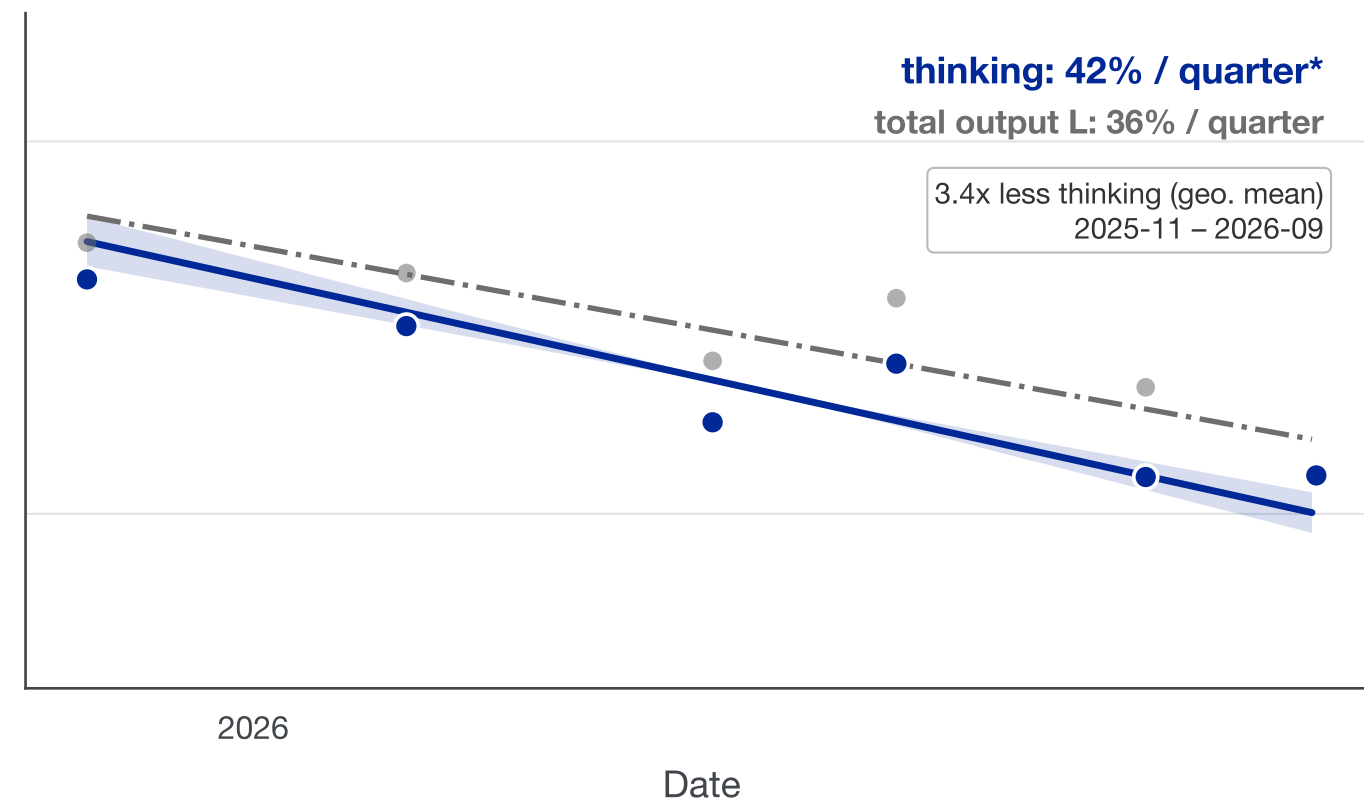
Robustness: the decline is in REASONING, not in answer formatting

Tokens per correct trace (geometric mean, log scale)

OpenAI (GPT)



Anthropic (Opus + Fable)



Thinking tokens (fit + models) 95% CI (wild cluster bootstrap) Total output L = thinking + answer

No floor is drawn: the minimal human derivation is a written derivation, the analogue of the model's ANSWER, so it does not bound thinking. Subtracting it would also censor 38% of gpt-6-astra's traces. Zero-thinking traces cannot enter log(thinking); all 81 are Fable 5.1, the newest model's shortest, so their loss makes the thinking-only rate conservative.